

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development
Module 1
Practical 7

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Class : TYIT	Batch : 1
Date of assignment : 07/08/2026	Date of assignment : 07/08/2026

4. Implement k-fold cross-validation on a classification model.

Code & Output:

Using iris.csv:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import cross_val_score
from sklearn.neighbors import KNeighborsClassifier

# Load dataset
iris = load_iris()

X = iris.data
y = iris.target

# Create classification model
model = KNeighborsClassifier()

# Apply 5-fold cross-validation
scores = cross_val_score(model, X, y, cv=5)

print("Accuracy scores for each fold:")
print(scores)

print("Average Accuracy:", round(scores.mean() * 100, 2), "%")
```

✓ 38.0s

Accuracy scores for each fold:
[0.96666667 1. 0.93333333 0.96666667 1.]
Average Accuracy: 97.33 %

Using wine.csv:

```

import pandas as pd
from sklearn.model_selection import cross_val_score
from sklearn.neighbors import KNeighborsClassifier

# Load dataset
wine = pd.read_csv("wine.csv")

# Features and Target
X = wine.iloc[:, :-1]
y = wine.iloc[:, -1]

# Create classification model
model = KNeighborsClassifier()

# Apply 5-fold cross-validation
scores = cross_val_score(model, X, y, cv=5)

print("Accuracy scores for each fold:")
print(scores)

print("Average Accuracy:", round(scores.mean() * 100, 2), "%")

```

✓ 0.4s

Accuracy scores for each fold:
[0.72222222 0.66666667 0.63888889 0.65714286 0.77142857]
Average Accuracy: 69.13 %

5. Implement a Decision Tree classifier using the Iris or any relevant dataset.

Code & Output:

Using iris.csv:

```

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

# Load dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=1
)

# Create model
model = DecisionTreeClassifier()

# Train model
model.fit(X_train, y_train)

# Display accuracy
print("Training Accuracy:", model.score(X_train, y_train) * 100)
print("Testing Accuracy:", model.score(X_test, y_test) * 100)

# Prediction
prediction = model.predict([X_test[0]])
print("Predicted Class:", prediction[0])

```

✓ 1.1s

```
Training Accuracy: 100.0
Testing Accuracy: 95.55555555555556
Predicted Class: 0
```

Using wine.csv:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

# Load dataset
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# Features and Target
X = wine.iloc[:, :-1]
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# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
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# Create model
model = DecisionTreeClassifier()

# Train model
model.fit(X_train, y_train)

# Display accuracy
print("Training Accuracy:", model.score(X_train, y_train) * 100)
print("Testing Accuracy:", model.score(X_test, y_test) * 100)

# Prediction
prediction = model.predict([X_test.iloc[0]])
print("Predicted Class:", prediction[0])
```

✓ 0.1s

```
Training Accuracy: 100.0
Testing Accuracy: 94.44444444444444
Predicted Class: 2
```

Justification

- **Different Features**

Iris has only 4 features (sepal and petal measurements).

Wine has 13 chemical features, making classification more complex.

- **Data Complexity**

The three Iris flower species are relatively easy to distinguish.

Wine classes have more features and some samples are similar, making classification harder.

- **Dataset Size**

Iris has 150 samples.

Wine has 178 samples.

A different number of samples affects model training and evaluation.

- **Different Data Distribution**

The values and relationships between features differ in each dataset.

Therefore, the classifier learns different decision boundaries.

- **Train-Test Split**

During Decision Tree classification, the training and testing data are split randomly (using `random_state=1`).

The selected samples are different for each dataset, so the testing accuracy also differs.